Huizhen (Serena) Sun

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Research Interests

Spatial data science, ocean colour remote sensing, and coastal resilience and climate adaptation.

Education

MSc in Geospatial Data Science, with Distinction Sep 2025 – Aug 2026
The University of Hong Kong *Hong Kong SAR, China*

- GPA: 3.61/4.3 Capstone supervisor: Prof. Hongsheng Zhang
- Relevant coursework: Satellite Data Processing and Information Extraction; Spatial Statistics; Cloud Computing for Geospatial Data Analytics; Geographic Information Systems

Bachelor of Management in E-Commerce Sep 2021 – Jun 2025
Xidian University *Xi'an, China*

- GPA: 3.6/4.0 (84%)
- Relevant coursework: Probability Theory and Mathematical Statistics; Applied Statistics; Big Data Analysis; R and Complex Data Analytics; Database Principles

Publications

Peer-Reviewed Journal Articles

[1] Zhao, D., Luo, Q., Lai, X., **Sun, H.**, Qiu, Z., Zhang, H., & Sun, Z. (2026). Normalized dynamic fluorescence height: An alternative algorithm for chlorophyll *a* estimation in algae-dominated waters using hyperspectral remote sensing reflectance from in situ and spaceborne imagers. *Remote Sensing, 18*(14), 2332. https://doi.org/10.3390/rs18142332

Manuscripts in Preparation

[2] **Sun, H.**, Zhao, D., & Sun, Z. (in preparation). Reconstructing gap-free daily total nitrogen and total phosphorus fields in small and medium-sized lakes from a multi-satellite virtual constellation: Futou Lake, China.

[3] **Sun, H.**, Zhao, D., Zhang, H., & Sun, Z. (in preparation). Decadal regime shifts and drivers in Xijiang River water quality.

Research Experience

MSc Capstone Project Jan 2026 – Aug 2026
Spatiotemporal Dynamics of Coastal Water Quality: The Hidden Role of Sea Level Rise *HKU*

- **Question:** Whether coastal morphology determines the direction of sea-level-rise effects on water quality in urbanised estuaries.
- **Data & methods:** 31 years of monthly data from 94 stations, combined with satellite altimetry, ocean and wave reanalysis, and coastal geophysical data; panel regression, causal mediation, and spatial autoregressive models in Python.
- **Findings:** A morphology-dependent sign reversal in the response across Hong Kong; stratification mediates 94% of the effect in enclosed embayments, robust to ENSO controls.

Multi-Satellite Chlorophyll-*a* Retrieval for Lake Taihu Sep 2025 – Dec 2025
Project Leader — Course GEOG7304, Satellite Data Processing and Information Extraction

- Led an 11-member team end to end: defined the research questions, planned the project, assigned tasks, and coordinated integration of results; awarded full marks.

- Designed a workflow harmonising Landsat 8/9, Sentinel-2, and GF-1 imagery to 30 m Landsat-equivalent R_{rs} (ACOLITE; band adjustment calibrated with in situ hyperspectral R_{rs}), with in situ data for training and GOCI data for validation.
- Built a Random Forest retrieval model ($R^2 = 0.810$, RMSE = 2.04 µg/L) that outperformed linear regression baselines, and produced multi-year, lake-wide chlorophyll-a maps.

Visiting Student, Ocean Colour Remote Sensing Sep 2024 – Jul 2025
School of Innovation and Entrepreneurship, SUSTech — Supervisor: Zhaohua Sun *Shenzhen, China*

- Trained in ocean colour science, marine optics, and the radiative properties of natural waters.
- Developed Python-based water quality retrieval algorithms from remote sensing reflectance and analysed the spatial distribution and temporal dynamics of water quality parameters.

Mathematical Contest in Modeling (MCM): Reimagine Masai Mara Feb 2023
Team project in remote sensing and environmental modelling

- Extracted land use, vegetation, and water information from remote sensing data for ecosystem modelling.
- Coupled ecosystem health and economic impact models through multi-objective optimisation to design and evaluate management strategies.

Professional Experience

Remote Sensing Engineer (Part-time) May 2026 – Aug 2026
AIEyes Technologies (Hong Kong) Co., Limited *Hong Kong SAR*

- Developed hyperspectral retrieval algorithm packages for 25+ water quality parameters (e.g., chlorophyll, CDOM, SPM), with an RMSE below 0.15.
- Contributed to a Hong Kong Science Park incubation program on AI-enabled water environment monitoring.

Data Engineer Assistant (Intern) Jul 2024 – Aug 2024
Anhua Ocean Intelligent Equipment (Shenzhen) Co., Ltd. *Shenzhen, China*

- Processed apparent optical property (AOP) data in Python to derive remote sensing reflectance (R_{rs}).
- Drafted requirements and webpage prototypes for a platform sharing AOP data with researchers.

Additional Training

Training Program in Machine Learning and NLP Apr 2024 – Jun 2024
Columbia University (online program)

- Applied machine learning and NLP (sentiment analysis, topic modelling) in Python to user reviews for location-based services.

Honors & Awards

Successful Participant, Mathematical Contest in Modeling (MCM) May 2024
Third-Class Scholarship, Xidian University Oct 2023
Provincial Third Prize, 14th "Zhengda Cup" National Market Research Competition Apr 2023

Skills

Programming	Python, R, MATLAB, Java, SQL (MySQL); intermediate C and LaTeX
Geospatial	ArcGIS, Google Earth Engine, GeoPandas
Statistics & ML	Random forest, panel regression, causal mediation analysis, spatial autoregressive models, clustering, NLP; SPSS
Languages	Chinese (native), English (fluent), German (intermediate)
Tests	IELTS 7.0 (Reading 8.0, Listening 7.5, Writing 6.5, Speaking 6.0), Jun 2024